Satellite Digital Twin for Detection of Falsified Spacecraft Telemetry Using NASA cFS and NOS3

COMP 8800 Major Project Proposal | BScACS – Network & Security, BCIT
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1. Problem Statement

Small satellites rely on telemetry from subsystems such as the Electrical Power System (EPS) and Attitude Determination and Control System (ADCS) to report spacecraft state. If that telemetry is falsified, replayed, or produced by a compromised component, a monitor that only inspects the reported values cannot tell whether they reflect the spacecraft's real physical state, especially when the forged values stay within normal operating ranges.

This project proposes an independent Digital Twin that estimates expected telemetry from physical models and independent inputs, then compares it with telemetry reported by a simulated satellite running NASA's Core Flight System (cFS) inside the NASA Operational Simulator for Small Satellites (NOS3). Large discrepancies (residuals) are flagged as possible telemetry manipulation.

2. Related Work and Gap

Digital Twin and model-based approaches already detect spacecraft *faults* by comparing observed and expected behavior: Shangguan et al. [1] for satellite power-system diagnosis, Djebko et al. [2] using residuals on SONATE housekeeping data, and Casado et al. [3] with a validated real-time 6U CubeSat EPS twin. NASA's Orion Digital Twin work [4] lists a "Sensor/Telemetry Integrity Sentinel" as a possible use case but does not evaluate it. NOS3 [5] provides an open cFS + 42 dynamics + COSMOS environment, and Fayyaz et al. [6] used it to build a CubeSat cybersecurity dataset.

Gap: prior twins target accidental faults and often synchronize with the very telemetry they check. No work found experimentally evaluates an *independent*, physics-based twin as a defence against *deliberately* falsified or replayed telemetry. This project tests that idea in cFS/NOS3.

3. Objectives

1. Set up a reproducible small-satellite environment (cFS, NOS3, 42, COSMOS).
2. Select a manageable set of telemetry parameters, starting with the EPS (battery, voltage, current, solar input, loads), with a dynamics-based parameter (position/attitude) as a second case if feasible.
3. Build twin models that predict each parameter *without* using the value being validated.
4. Compute residuals and set detection thresholds from a normal-operation baseline.
5. Inject controlled attacks and measure detection rate, false-positive rate, latency, and overhead.

4. Methodology

The system runs as a monitoring layer beside the simulator with two paths (Fig. 1): the *reported* state (NOS3 → cFS → telemetry) and the *expected* state (twin models using dynamics, known commands, prior state, and environment). For each monitored parameter:

$$r(t) = x_{\text{reported}}(t) - \hat{x}_{\text{twin}}(t)$$

An alert is raised when $|r(t)|$ exceeds a baseline-derived threshold or a physical/temporal consistency rule is violated. An alert means "inconsistent with expected state"; it does not by itself attribute the cause (attack, fault, or model error).

Experiments (simulation only):

- **Baseline:** no manipulation; characterize normal residuals and false positives.

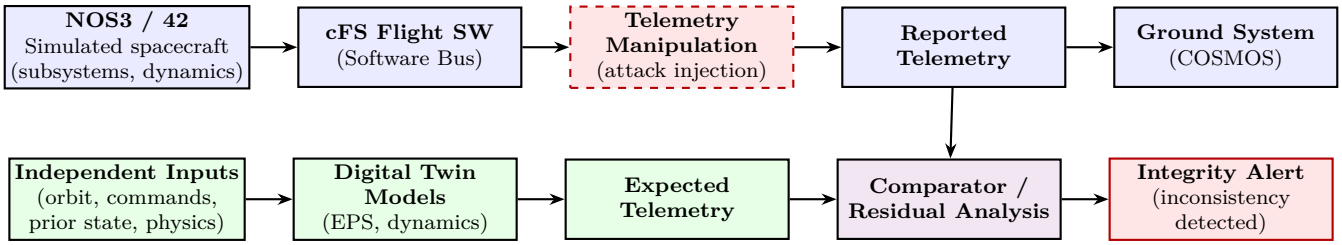


Figure 1: Proposed Digital Twin-based telemetry-integrity architecture: reported path (top) vs. independently estimated path (bottom).

- **Value falsification:** alter reported values while the simulated physical state is unchanged, including in-range forgeries.
- **Replay/stale telemetry:** replay old valid values while the true state keeps evolving.
- **Sensitivity:** vary manipulation magnitude to find detection limits.

Metrics: detection rate, false-positive rate, detection latency, residual magnitude, computational overhead.

Scope: not a full-spacecraft twin, a replacement for link encryption, or a complete IDS; the modular design allows more subsystems later.

5. Deliverables and Success Criteria

Deliverables: configured NOS3/cFS environment; Digital Twin prototype; telemetry-integrity monitor (residuals + alerts); manipulation test framework; experimental evaluation; source code, architecture diagram, final report, and demo.

Success: repeatable telemetry and attack injection; the twin independently predicts selected parameters; a measurable share of injected inconsistencies is detected with a reported false-positive rate, latency, and overhead; and results clearly show both strengths and limitations.

6. Key Risks

- **NOS3 limitations:** some simulated components/interfaces are unreliable (an early radio-based design was dropped). *Mitigation:* pick telemetry based on what NOS3 exposes reliably.
- **Model accuracy and independence:** simple models cause baseline residuals, and reusing the checked value causes circular validation. *Mitigation:* baseline-calibrated thresholds; document each model's independent inputs.
- **Scope creep:** *Mitigation:* limit to EPS.

7. Contribution

The project moves Digital Twins from fault diagnosis to **cyber-physical telemetry integrity**, emphasizing model independence, in an open, reproducible cFS/NOS3 testbed. It extends my Network & Security specialization to flight software and cyber-physical systems.

References

[1] Y. Shangguan, L. Chen, and J. Ding, "A Digital Twin-Based Approach for the Fault Diagnosis and Health Monitoring of a Complex Satellite System," *Symmetry*, vol. 12, no. 8, Art. 1307, 2020.

[2] S. Djebko, F. Puppe, and H. Kayal, "Model-Based Fault Detection and Diagnosis for Spacecraft with an Application for the SONATE Triple Cube Nano-Satellite," *Aerospace*, vol. 6, no. 10, Art. 105, 2019.

[3] P. Casado, C. Torres, J. M. Blanes, A. Garrigós, and D. Marroquí, "Implementation of a 6U CubeSat Electrical Power System Digital Twin," *Aerospace*, vol. 11, no. 8, Art. 688, 2024.

[4] G. J. Pierce, J. D. Heeren, and T. R. Hill, "Orion SysML Model, Digital Twin, and Lessons Learned for Artemis I," NASA Johnson Space Center, Houston, TX, USA.

[5] J. P. Lucas *et al.*, "NASA Operational Simulator for SmallSats (NOS3) – Design Reference Mission," in *Proc. 37th Annual Small Satellite Conf.*, SSC23-XIII-06, 2023.

[6] Y. Fayyaz *et al.*, "CubeSat Cybersecurity Dataset for Intrusion Detection (CuCD-ID): Labelled NOS3/cFS Telemetry (Raw + Augmented) with COSMOS Reproduction Scripts," *Data in Brief*, vol. 65, Art. 112598, 2026.